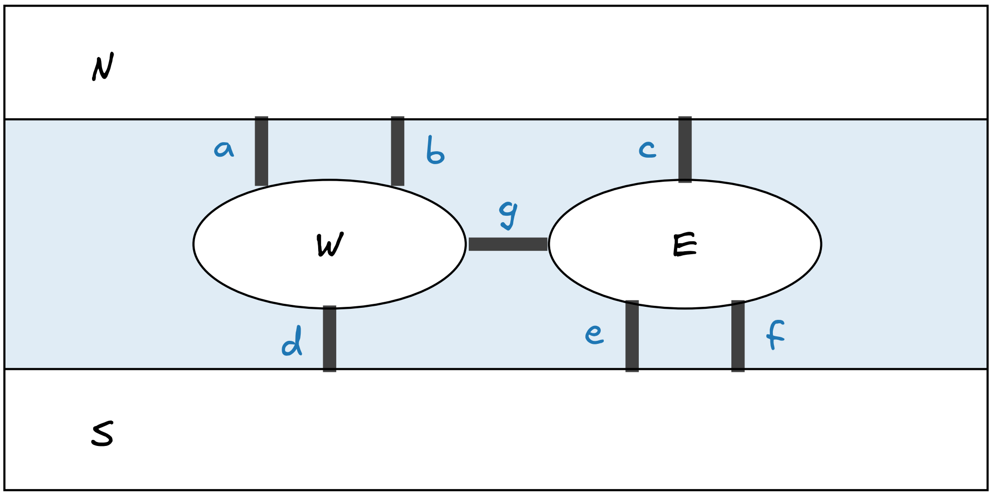
Cross Every Bridge Once, or Prove Nobody Can

You have a summer job as a campus tour guide. The campus sits on both banks of a creek, with two small islands in the middle, all stitched together by footbridges. Every footbridge carries a donor's name on a brass plaque, and the development office wants a walking tour that crosses every footbridge exactly once, so that no donor is left out.

Start anywhere, finish anywhere, set foot on the same landmass as often as you like. Do not cross the same footbridge twice, and do not swim.



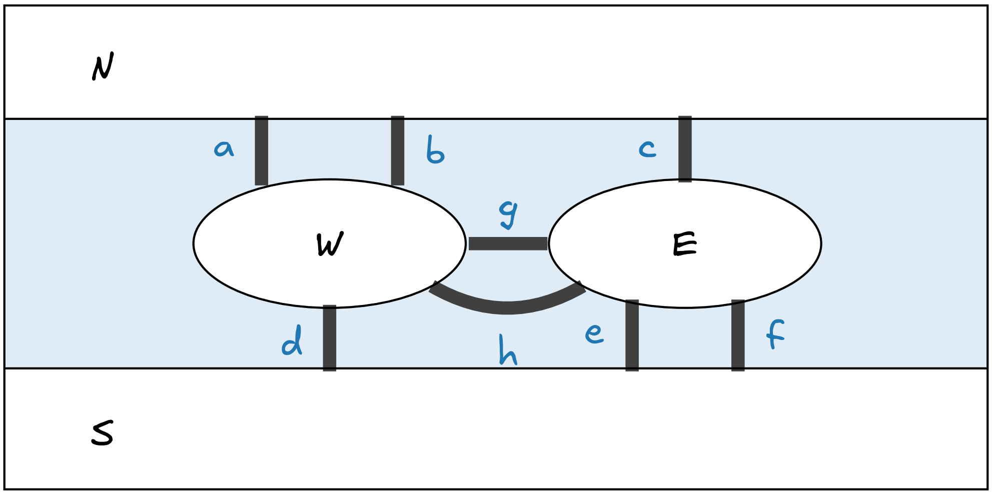
N = north bank, S = south bank, W = west island, E = east island. The seven footbridges are a through g.

Question 1: Find a tour. Write it as the sequence of bridge letters you cross. Cross your failed attempts out rather than erasing them — you will want to look at them again.

Start at ________ then cross ______________________________

Room to try, and to cross out:

Question 2: The development office finds one more donor and builds an eighth footbridge, h, between the two islands.



Before you touch it: does an extra bridge make the tour easier or harder to find? Circle one and say why in one sentence. Keep this page — you will be asked about your answer again at the end.

easier harder no difference

because ________________________________

Now try it. Give yourself three honest attempts. Each time, write down which bridge you were left holding and where you were standing when you ran out.

Question 3: Throw the map away. Redraw the seven-bridge campus (Map A, no bridge h) using only four dots and seven lines. Then draw it a second time with the four dots in completely different positions — put W above N if you like.

Is the tour question the same for both drawings? What did you throw away when you replaced the map with dots and lines, and how do you know that throwing it away was safe?

Question 4: Keep the books. Take the tour you found in Question 1. Walk it again slowly, and at each landmass count two things separately: how many times you arrived there by crossing a bridge, and how many times you left it by crossing a bridge. Then count how many bridges touch that landmass in the first place.

Landmass	arrived	left	arrived + left	bridges here
N				
W				
S				
E				

Two of these columns came out identical. Which two, and why did they have to?

Question 5: Look down the table. At exactly two landmasses, "arrived" and "left" disagree. Name them, and say what those two landmasses were doing in your tour.

Now finish these two sentences, using the last column of the table.

A landmass you only ever pass through must have an ___________ number of bridges, because

____

A landmass with an odd number of bridges can only be

____

Question 6: Go back to Map B, the eight-bridge campus. Without walking a single route, write down how many bridges touch each landmass on each map, and decide the case.

	N	W	S	E
Map A				
Map B				

A tour has one beginning and one end. So at most _____ landmasses may have an odd number of bridges. Map A has _____, and Map B has _____.

Write the verdict on Map B as a sentence starting "No tour can exist, because..." — not "I could not find one".

Question 7: The number of bridges touching a landmass is its degree, and a tour crossing every bridge exactly once is an Eulerian trail. You have just derived the test for one. State it.

A campus has an Eulerian trail when

Now the development office adds a condition: the tour must end back at the food truck it started from. What changes in the sentence you just wrote, and why? Answer with the "arrived / left" bookkeeping, not by trying routes.

Question 8: Look at your Map A tour once more and answer two small questions about it.

Did you cross any bridge twice? _______ Did you set foot on any landmass twice? _______

Three names for three levels of strictness. A walk is any sequence of landmasses joined by bridges — repeat whatever you like. A trail never repeats a bridge. A path never repeats a landmass. Close any of them up so that it ends where it began and you get a circuit (a closed trail) or a cycle (a closed path).

Which of the three is your Map A tour? ______________

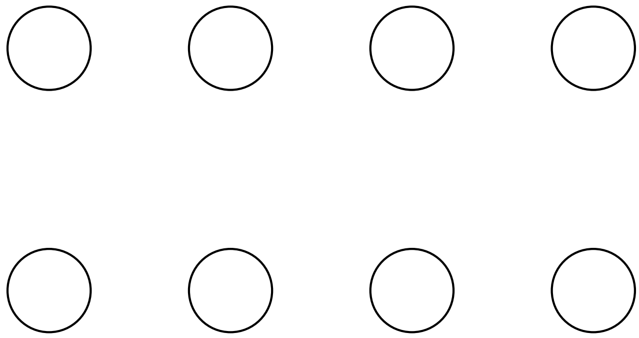
Could a route crossing all seven of Map A's bridges ever be a path? Answer with a count, not with an attempt.

Question 9: Add up the degrees of all four landmasses of Map A, and write the total here: _________

Compare it with the number of bridges, and explain the relationship in one sentence, in terms of what every bridge has two of.

Now use it. Design a campus with exactly three landmasses of odd degree. Try hard, then report what happened and why.

Question 10: Break your own rule. Below are eight blank land-masses. Draw bridges so that every landmass has an even degree, and yet no tour crossing every bridge exactly once exists.



Your answer to Question 7 promised such a tour would exist. Go back and add the clause you were missing. Write it here too.

Question 11: Repair the campus. Map B has no tour. You have two ways to fix it: build new footbridges, or close existing ones.

Fewest new bridges to build: _______

between which landmasses? _________________________

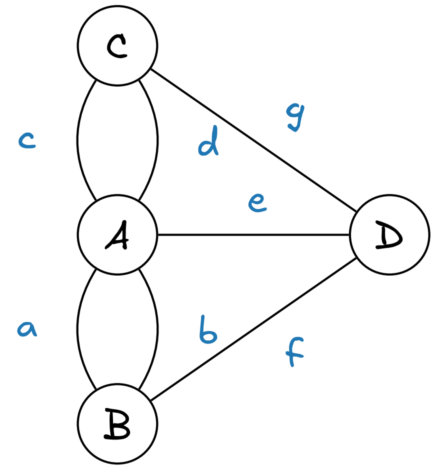
Fewest existing bridges to close: _______

which ones? ______________________________

Both answers came out the same size. Say what a single bridge does to the count of odd-degree landmasses, and why that explains it.

A campus is all one piece and has $2k$ landmasses of odd degree. What is the fewest new bridges needed to make a tour possible? Give the number in terms of k, and justify it.

Question 12: Königsberg, 1736. Here is a real city, already stripped to dots and lines: four landmasses, seven bridges. Two pairs of bridges run between the same landmasses, and each bridge in a pair counts separately, because each one has to be crossed.



The citizens of Königsberg spent decades hunting for a Sunday walk crossing all seven exactly once. Settle it with your rule instead, in under a minute.

Degrees: A ____ B ____ C ____ D ____

Verdict: ______________________________

In 1944 the city was bombed, two of these bridges were destroyed, and the walk became possible. Which single bridge, destroyed on its own, would already have been enough? List every bridge that works.

Your list should be suspiciously long. What does that tell you about this particular city that the verdict alone did not?

Question 13: Back in Question 2 you guessed whether an eighth bridge makes the tour easier or harder. Were you right? Say in one sentence what building a bridge actually does — not to the number of possible routes, but to the thing your rule looks at.

Now write the recipe you would hand to a computer so that it can decide, for a city with a million landmasses and three million bridges, whether the tour exists. No calculation — just say what it has to count, and what it has to check that is not a count.

Question 14: Change one word in the assignment. The tour must now visit every landmass exactly once, instead of crossing every bridge exactly once.

Find such a route on Map A. ______________________________

Does the degree rule you built in Question 7 predict whether one exists? Try to state a rule of the same kind — "count the bridges at each landmass, then..." — that decides this new question.

If you cannot, say precisely where the bookkeeping of Question 4 stops working. That failure is not yours: nobody has such a rule, and the two questions differ by a single word.